Prohori: Progress Report

OpenCV AI Competition 2026 — AWS Compute Grant check-in

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1 Project

Prohori measures concrete crack width in millimetres from a phone photo. The user places a printed ChArUco board next to the crack; OpenCV uses the board to correct for camera angle and scale and reports the width with an uncertainty range. When a photo cannot support a reliable measurement, the app tells the user how to retake it.

Live app: `https://184-194-241-186.sslip.io` (tap *Demo samples* to try it without a board).

2 Progress

- **Measurement pipeline:** ChArUco detection, `solvePnP` pose, rectification to a known px/mm, crack location, and per-crack width with uncertainty (about 1.5 s per 12 MP photo).
- **Quality gate:** a learned check, calibrated with conformal risk control, that certifies a width only when the photo supports it, and attaches a 90% range to each width.
- **Capture agent:** after each photo, an Amazon Bedrock agent (Nova Lite) chooses the next step (accept, move closer, hold steadier, fix lighting, reposition the board, or stop) from OpenCV's measurements. A free rules policy is the default.
- **Phone web app, history and alerts:** per-joint visit history, change detection beyond measurement uncertainty, engineer alerts with an approval step, and automatic face blurring (OpenCV YuNet) before any photo is stored.
- **Testing:** 206 automated tests passing.

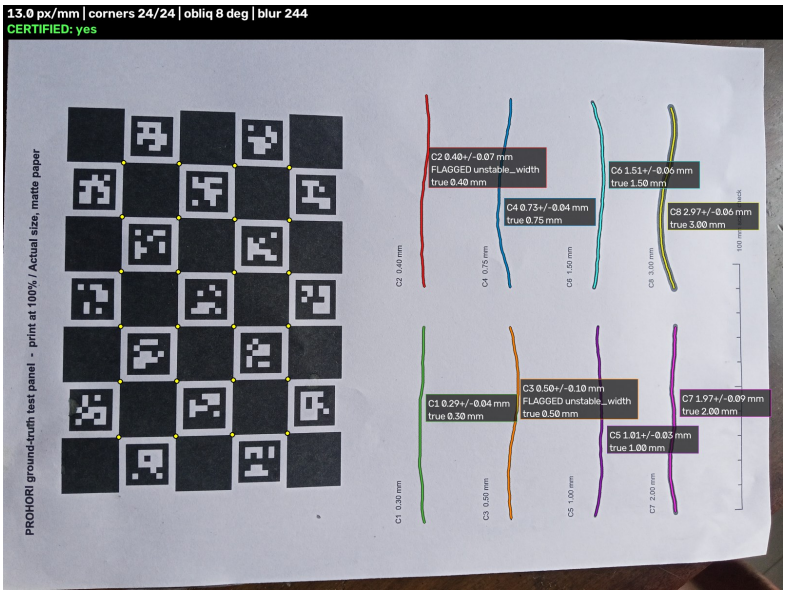


Figure 1: A phone photo processed by Prohori. The board corners set the scale; each crack is labelled with its measured width next to the true printed width (e.g. 0.29 mm vs 0.30, 1.01 mm vs 1.00, 2.97 mm vs 3.00).

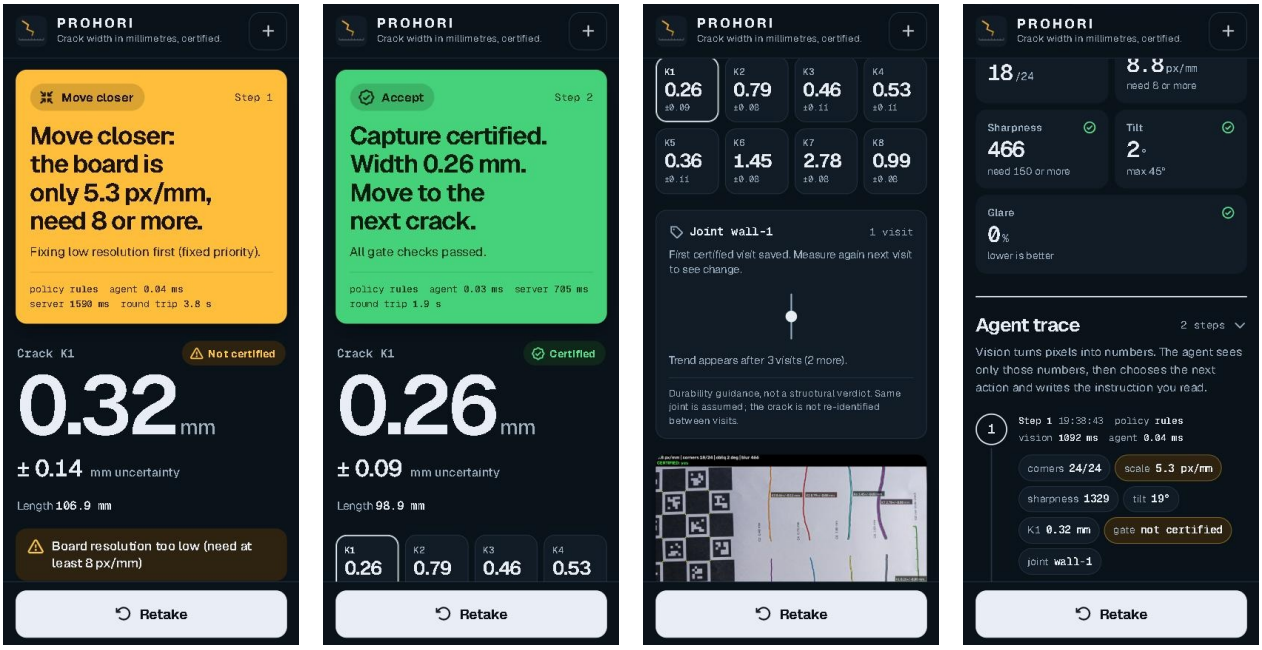


Figure 2: The live phone app. Left to right: photo taken too far away (move closer); certified width after moving closer; all cracks and joint history; photo quality and agent trace.

3 Results so far

Tested on 29 phone photos of a printed panel with eight cracks of known width (0.30–3.00 mm):

Average width error, certified cracks (held-out photos)	0.048 mm
True width inside the stated 90% range (held-out photos)	92.9%
Capture agent recovery on multi-problem photos (vs fixed rules)	85% (62%)

4 AWS usage

Prohori runs as a production service in `us-east-1`:

- **EC2 Graviton t4g.small** running OpenCV 5.0 behind HTTPS (Elastic IP, Caddy).
- **Amazon Bedrock** (Nova Lite) for the capture agent, capped at 200 calls per day.
- **DynamoDB** for agent traces and visit history; **S3** (private, encrypted) for photos.
- **SNS** for engineer alerts; **IAM**, **SSM** and an **AWS Budgets** alarm.

The service was built to run at low cost, so usage to date is covered by the account's Free Tier credits ($120 remaining on 19 October); the full Bedrock agent evaluation cost $0.06. The grant funds the final stage:

Item	Estimate
GPU training of a learned crack-segmentation model (EC2 GPU instance)	$25
Graviton4 (c8g) benchmark with the OpenCV COOL build (Paid plan)	$20
Hosting the live service through judging and a 3-month pilot	$55
Bedrock usage for evaluation, demo video and judging	$10
Storage, alerts and field-test traffic	$40
Total	**$150**

5 Remaining work (to 26 October)

Deploy the calibrated quality gate and updated agent	21 Oct
Train the learned crack-segmentation model	23 Oct
Graviton4 + OpenCV COOL benchmark	24 Oct
Improve thin-crack width accuracy	24 Oct
Validation on real walls in Dhaka	25 Oct
Write-up, demo video and submission	26 Oct

Thank you for the grant and for the flexibility on the check-in.